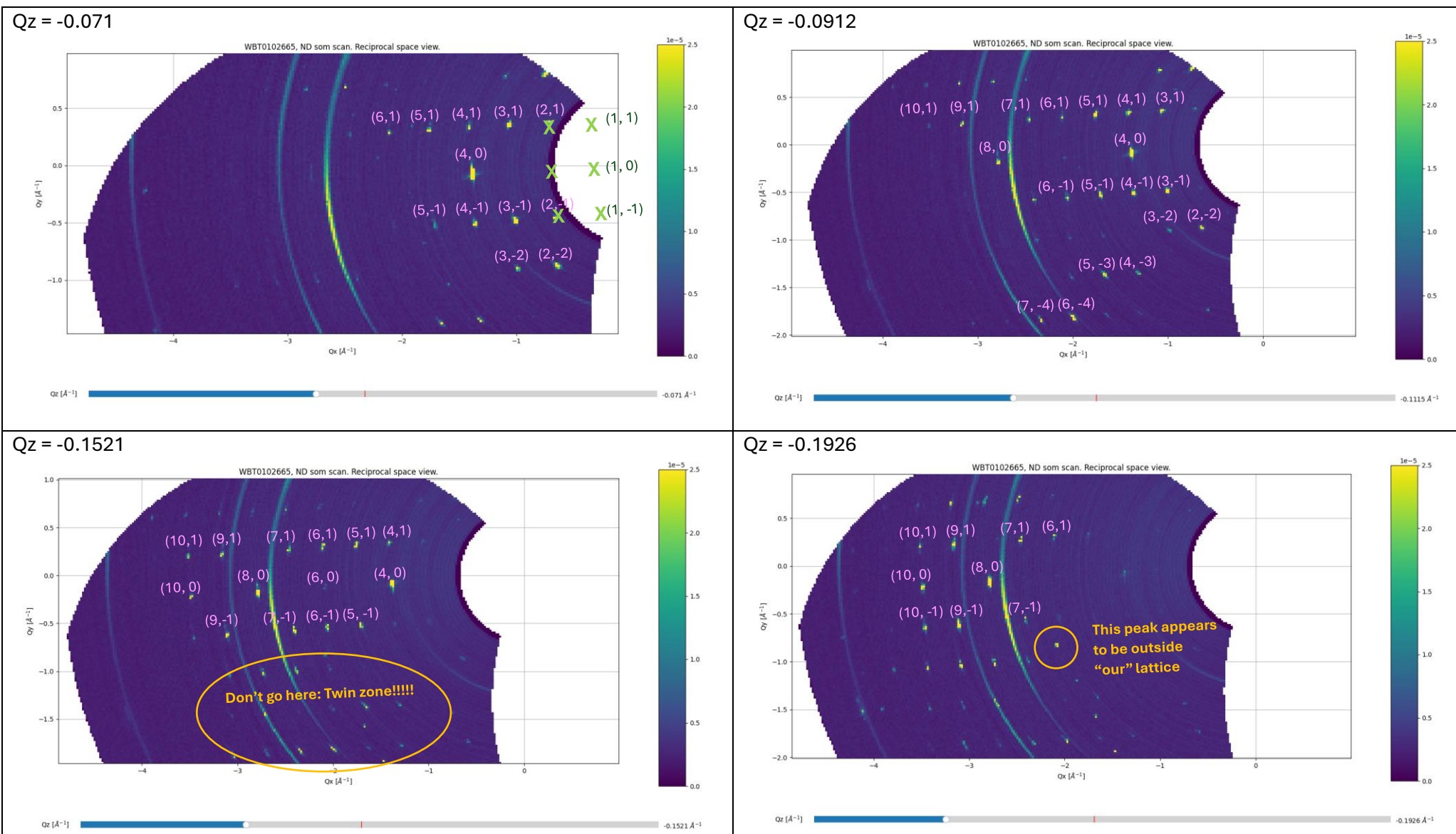
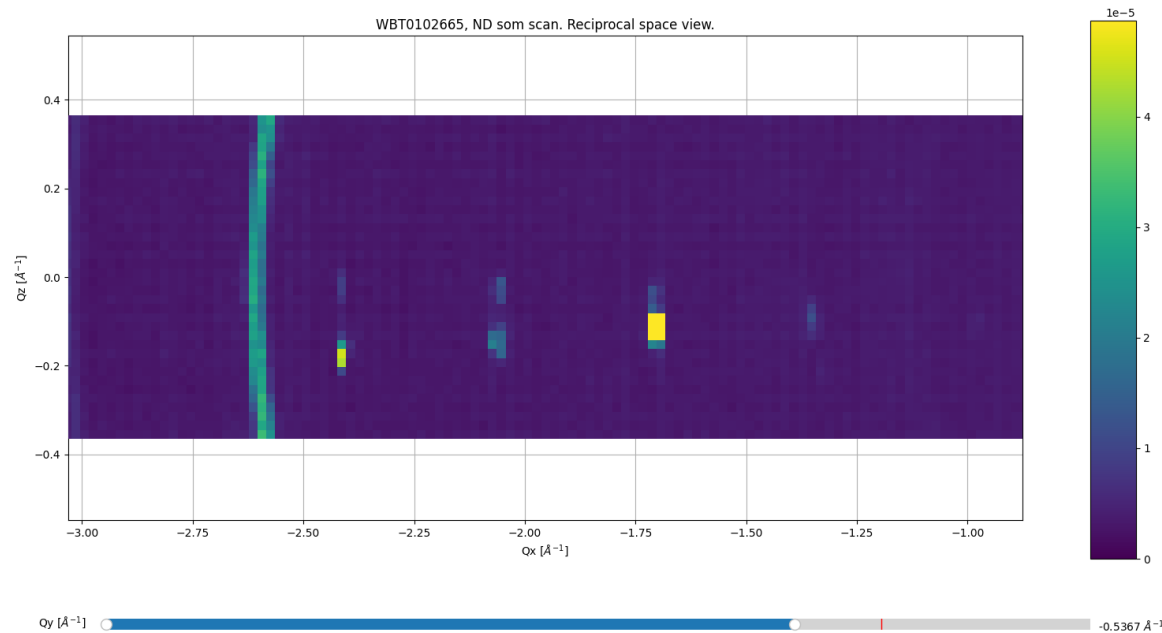


Indexing Alex's crystal

1. Looked at Qx-Qy plane (scattering plane of detector). Could see a lattice of spots, depending on Qz ($-0.1926 \text{ \AA}^{-1} \leq Qz \leq -0.071 \text{ \AA}^{-1}$ shown below), different parts of the lattice got brighter, suggesting there was a scattering plane nearly parallel to the scattering plane of the detector
2. Indexed spots with just two indices – some of the low Q spots were inaccessible (indicated by green X below)



3. Got co-ordinates (Q_x , Q_y , Q_z) of a few of these reflections and added to spreadsheet based on positions in Q_x - Q_y view shown above.
4. Looked at these reflections AGAIN but in orthogonal Q_x - Q_z plane, to refine Q_z co-ordinate of spots. Where twinning was apparent peaks seemed to split in Q_z , I always chose the lower, brighter twin (e.g. below).



5. For UB matrix calculation, we need two spots in (Q_x , Q_y , Q_z) and indexed hkl ... I already had two out of three indices I assigned above simply based on counting reflections. To get the third index, I looked at the list of calculated structure factors and compared 2θ , then made a choice based on intensity and hkl . Remember the 2θ of the observed reflection is calculated from (Q_x , Q_y , Q_z) which just comes from the position of my mouse where I think the peak is brightest – so a little inaccurate!

Handy dandy reciprocal space conversion thing					wavelength (AA)										
					2.41										
Qx (AA^-1)	Qy (AA^-1)	Qz (AA^-1)	Q (AA^-1)	d (AA)	2θ (deg)	Notes	Candidate peaks based on 2θ (ordered by intensi	By hand Qx index	By hand Qy index	Col	h	k	l		
-0.714	0.376	-0.08	0.81090813	7.74833184	17.89360364	Twinned - chose peak lower in Qz (stronger),	(1 0 0), (0 0 2), (0 2 1) -- (0 2 1) is weakish, which fits	2	1		0	2	1		
-1.061	0.347	-0.075	1.11881857	5.6159108	24.78047783	Twinned - chose peak lower in Qz (stronger)	(1 1 1), (0 2 2), (0, 3, 1) -- very similar intensities	3	1		0	3	1		
-1.414	0.347	-0.089	1.45867268	4.30746759	32.49017948	Twinned - chose peak lower in Qz (stronger)	(0 4 1), (1 2 -3)	4	1		0	4	1		
-1.384	-0.087	-0.087	1.38945817	4.52203992	30.90895356	? Twinning begins	(1 0 2), (0 4 0)	4	0		0	4	0		
-2.094	-0.136	-0.129	2.10237318	2.98861561	47.55634677	weak	(0 6 0) -- weakish	6	0		0	6	0		
-2.787	-0.178	-0.171	2.79790886	2.2456719	64.90330391		(0 8 0)-- moderately strong	8	0		0	8	0		
-1.009	-0.473	-0.078	1.11709176	5.62459194	24.74162489		(0 3 1) or (0 -3 -1) or (0 -3 1) or (0 3 -1)	3	-1		0	3	-1		
-1.362	-0.508	-0.096	1.45681982	4.31294605	32.44776888		(0 4 1) M4	4	-1		0	4	-1		
-3.1105	-0.6192	-0.191	3.17727869	1.97753673	75.08415979	Twinned - chose peak lower in Qz (stronger)	(0 9 1) M4	9	-1		0	9	-1		

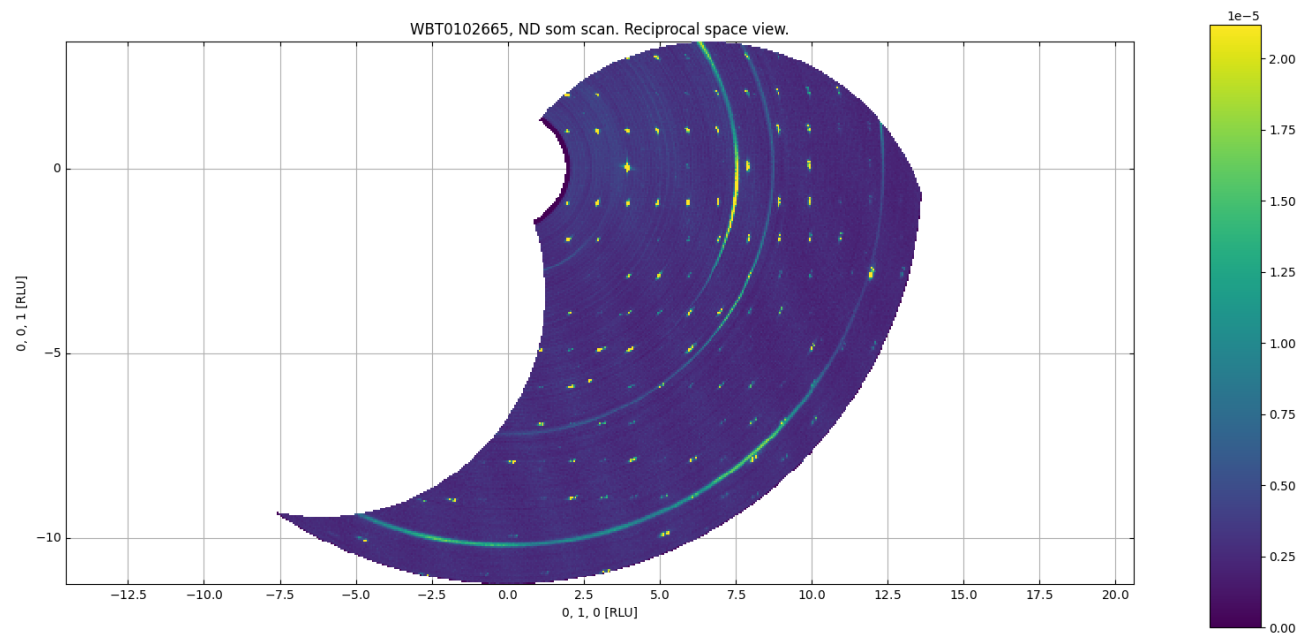
6. For example, the peak at $2\theta = 17.89^\circ$ could be (1 0 0), (0 0 2), or (0 2 1). Given I already had “2” and “1” identified by hand, I chose (0 2 1), calculated $2\theta = 18.2^\circ$. This peak also wasn’t super strong in the Wombat data, whereas (0 0 2) is supposed to be quite a strong reflection.

	A	B	C	D	E	F	G	H	I	J
	h	k	l	d (Å)	F(real)	F(imag)	F	2θ	I	M
2	0	1	1	11.302807	-1.374481	0	1.37448	12.23995	0.02092	4
3	0	2	0	8.9	111.105587	0	111.106	15.56272	42.52403	2
4	1	0	0	7.643794	119.328524	0	119.329	18.14037	36.30248	2
5	0	2	1	7.603721	-34.954291	0	34.9543	18.23679	6.16555	4
6	0	0	2	7.315534	78.472183	0	78.4722	18.9617	14.39666	2
7	1	1	-1	7.305216	10.641362	0	10.6414	18.98873	0.52801	4

Another example, the peak at $2\theta = 30.9^\circ$ could be (0 3 2), (1 0 2), (0 4 0) or (1 1 2). In the Wombat data, this spot is the brightest of them all, so that narrows it down to (1 0 2) or (0 4 0). Based on my counting of spots in the lattice, I went with (0 4 0), calculated $2\theta = 31.4^\circ$.

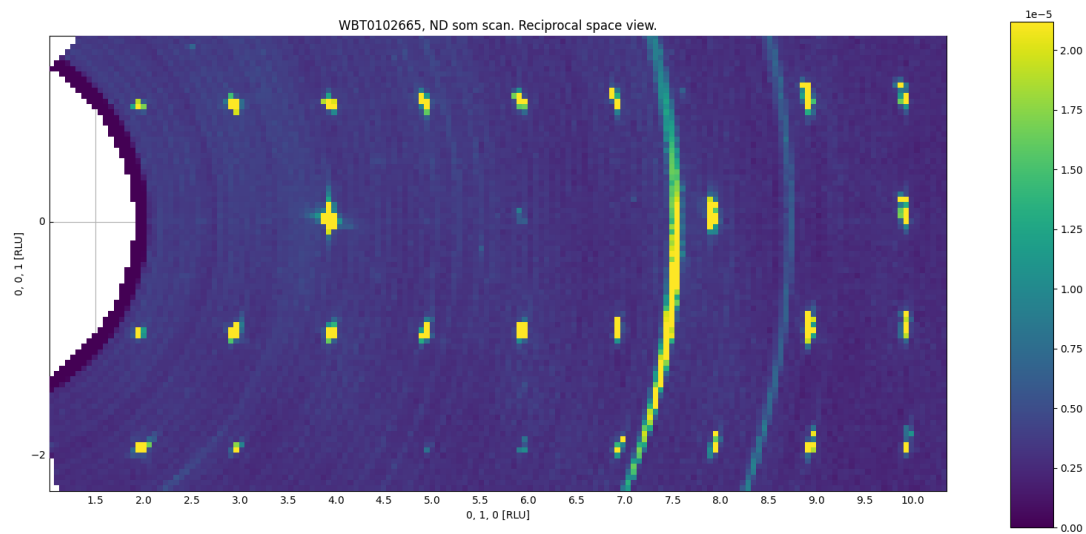
	A	B	C	D	E	F	G	H	I	J
	h	k	l	d (Å)	F(real)	F(imag)	F	2θ	I	M
3	0	3	2	4.60819	-35.437665	0	35.4377	30.3171	2.38104	4
4	1	0	2	4.554428	191.146401	0	191.146	30.68372	33.86287	2
5	0	4	0	4.45	215.962871	0	215.963	31.42215	41.34087	2
6	1	1	2	4.412286	27.079261	0	27.0793	31.69777	1.27887	4

7. Now choosing peaks for the UB matrix calculation:
- We probably only have access to one scattering plane (unless using the Eulerian cradle).
 - We want peaks with not too many zeroes in their hkl , i.e. (0 4 0) would be a bad choice because it doesn’t constrain h or l really at all
 - Ideally we want as many different “numbers” in the hkl , which is a bit constrained by point a.
 - We want peaks that are not on the same line, i.e. (0 1 2) and (0 2 4) together would be bad choices
 - We want peaks that are separated in Q but especially separated in Q_z – having a difference in the Q_z is important to constrain the tilt of the scattering plane.
 - Looking at the peaks in my list, I deliberately went and found the (0 9 -1) peak ($Q_z = -0.191 \text{ \AA}^{-1}$) as all the other peaks were between $Q_z = -0.08 \text{ \AA}^{-1}$ and $-0.129 \text{ \AA}^{-1}$ ignoring (0 8 0) as per point b. above.
8. I chose (0 3 1) and (0 9 -1):
- Five different numbers in the hkl – only repeat is one 0, which can’t be helped
 - Not on the same line
 - Separated in Q_z by $-0.183 \text{ \AA}^{-1}$ and separated in Q by $2 \text{ \AA}^{-1}$
 - Peaks chosen were from the same twin as far as I could tell.
9. Ran code – seemed to get a good result – bulk peaks are in the (0 k l) plane!!!
10. When we get Int3D, Int3D will find the centres of peaks in (Q_x , Q_y , Q_z) which will increase the accuracy of the indexing++ (and it can automatically find a UB matrix + refine the unit cell from the peaks it has found).



H: +0.197
K: +0.000
L: -0.131

0.0653 RLU



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